Alan Grafen pointed out to me another and more worrying problem with the account of hymenopteran sex ratios given in the first edition of this book. I have explained his point in *The Extended Phenotype* (pp. 75–6). Here is a brief extract:

> The potential worker is *still* indifferent between rearing siblings and rearing offspring at any conceivable population sex ratio. Thus suppose the population sex ratio is female-biased, even suppose it conforms to Trivers and Hare's predicted 3:1. Since the worker is more closely related to her sister than to her brother or her offspring of either sex, it might seem that she would 'prefer' to rear siblings over offspring given such a female-biased sex ratio: is she not gaining mostly valuable sisters (plus only a few relatively worthless brothers) when she opts for siblings? But this reasoning neglects the relatively great reproductive value of males in such a population as a consequence of their rarity. The worker may not be closely related to each of her brothers, but if males are rare in the population as a whole each one of those brothers is correspondingly highly likely to be an ancestor of future generations.

p. 242 If a population arrives at an ESS that drives it extinct, then it goes extinct, and that is just too bad.

The distinguished philosopher the late J. L. Mackie has drawn attention to an interesting consequence of the fact that populations of my 'cheats' and 'grudgers' can be simultaneously stable. 'Just too bad' it may be if a population arrives at an ESS that drives it extinct; Mackie makes the additional point that some kinds of ESS are more likely to drive a population extinct than others. In this particular example, both Cheat and Grudger are evolutionarily stable: a population may stabilize at the Cheat equilibrium or at the Grudger equilibrium. Mackie's point is that populations that happen to stabilize at the Cheat equilibrium will be more likely subsequently to go extinct. There can therefore be a kind of higher-level, 'between ESS', selection in favour of reciprocal altruism. This can be developed into an argument in favour of a kind of group selection that, unlike most theories of group selection, might actually work. I have spelled out the argument in my paper 'In Defence of Selfish Genes'.